

CSS Media Queries

CSS2 Introduced Media Types

- The **@media** rule, introduced in CSS2, made it possible to define different style rules for different media types.
- Examples: You could have one set of style rules for computer screens, one for printers, one for handheld devices, one for television-type devices, and so on.
- Unfortunately these media types never got a lot of support by devices, other than the print media type.

CSS3 Introduced Media Queries

- Media queries in CSS3 extended the CSS2 media types idea: Instead of looking for a type of device, they look at the capability of the device.
- Media queries can be used to check many things, such as:
 - width and height of the viewport
 - width and height of the device
 - orientation (is the tablet/phone in landscape or portrait mode?)
 - resolution
- Using media queries are a popular technique for delivering a tailored style sheet to desktops, laptops, tablets, and mobile phones (such as iPhone and Android phones)

Media Query Syntax

- A media query consists of a media type and can contain one or more expressions, which resolve to either true or false.

```
@media not|only mediatype and (expressions) {  
    CSS-Code;  
}
```

- The result of the query is true if the specified media type matches the type of device the document is being displayed on and all expressions in the media query are true. When a media query is true, the corresponding style sheet or style rules are applied, following the normal cascading rules.
- Unless you use the not or only operators, the media type is optional and the all type will be implied.

CSS3 Media Types

Value	Description
all	Used for all media type devices
print	Used for printers
screen	Used for computer screens, tablets, smart-phones etc.
speech	Used for screenreaders that "reads" the page out loud

Media Queries Simple Examples

- One way to use media queries is to have an alternate CSS section right inside your style sheet.
- The following example changes the background-color to lightgreen if the viewport is 480 pixels wide or wider (if the viewport is less than 480 pixels, the background-color will be pink)

```
@media screen and (min-width: 480px) {  
  body {  
    background-color: lightgreen;  
  }  
}
```

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CSS Media Queries - Examples

CSS Media Queries - More Examples

- Let us look at some more examples of using media queries.
- Media queries are a popular technique for delivering a tailored style sheet to different devices. To demonstrate a simple example, we can change the background color for different devices

- https://www.w3schools.com/css/tryit.asp?filename=trycss_mediaqueries_ex1

```
/* Set the background color of body to tan */
body {
    background-color: tan;
}

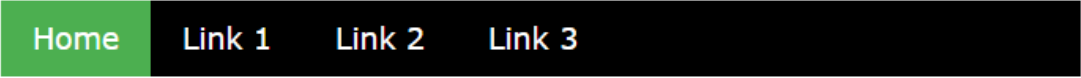
/* On screens that are 992px or less, set the background color to blue */
@media screen and (max-width: 992px) {
    body {
        background-color: blue;
    }
}

/* On screens that are 600px or less, set the background color to olive */
@media screen and (max-width: 600px) {
    body {
        background-color: olive;
    }
}
```

Media Queries For Menus

- In this example, we use media queries to create a responsive navigation menu, that varies in design on different screen sizes.

Large screens:

A horizontal navigation bar on a large screen. It consists of a dark blue bar with four items: 'Home' (highlighted in light blue), 'Link 1', 'Link 2', and 'Link 3'.

Small screens:

A vertical navigation menu on a small screen. It consists of a light blue bar with 'Home' at the top, followed by a dark blue bar containing 'Link 1', 'Link 2', and 'Link 3' stacked vertically.

- [Try it Yourself »](#)

```
/* The navbar container */
.topnav {
  overflow: hidden;
  background-color: #333;
}

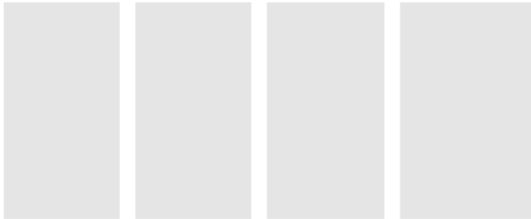
/* Navbar links */
.topnav a {
  float: left;
  display: block;
  color: white;
  text-align: center;
  padding: 14px 16px;
  text-decoration: none;
}

/* On screens that are 600px wide or less,
each other */
@media screen and (max-width: 600px) {
  .topnav a {
    float: none;
    width: 100%;
  }
}
```

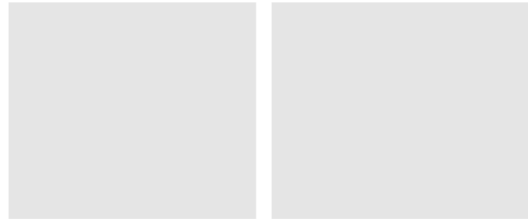
Media Queries For Columns

- A common use of media queries, is to create a flexible layout. In this example, we create a layout that varies between four, two and full-width columns, depending on different screen sizes

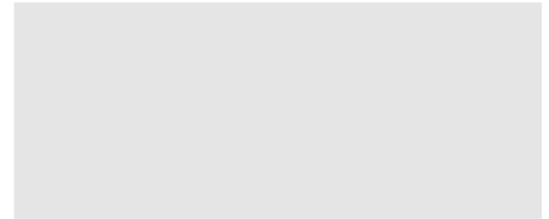
Large screens:



Medium screens:



Small screens:



- [Try it Yourself »](#)

```
/* Create four equal columns that floats next to each other */
.column {
    float: left;
    width: 25%;
}

/* On screens that are 992px wide or less, go from four columns to two columns */
@media screen and (max-width: 992px) {
    .column {
        width: 50%;
    }
}

/* On screens that are 600px wide or less, make the columns stack on top of each other instead of next to each other */
@media screen and (max-width: 600px) {
    .column {
        width: 100%;
    }
}
```

```
/* Container for flexboxes */
.row {
  display: flex;
  flex-wrap: wrap;
}

/* Create four equal columns */
.column {
  flex: 25%;
  padding: 20px;
}

/* On screens that are 992px wide or less, go from four columns to two columns */
@media screen and (max-width: 992px) {
  .column {
    flex: 50%;
  }
}

/* On screens that are 600px wide or less, make the columns stack on top of each other instead of next to each other */
@media screen and (max-width: 600px) {
  .row {
    flex-direction: column;
  }
}
```

• [Try it Yourself »](#)

Hide Elements With Media Queries

- Another common use of media queries, is to hide elements on different screen sizes

```
/* If the screen size is 600px wide or less, hide the element */  
@media screen and (max-width: 600px) {  
  div.example {  
    display: none;  
  }  
}
```

- [Try it Yourself »](#)

Change Font Size With Media Queries

- You can also use media queries to change the font size of an element on different screen sizes

```
/* If screen size is more than 600px wide, set the font-size of <div> to 80px */
@media screen and (min-width: 600px) {
  div.example {
    font-size: 80px;
  }
}

/* If screen size is 600px wide, or less, set the font-size of <div> to 30px */
@media screen and (max-width: 600px) {
  div.example {
    font-size: 30px;
  }
}
```

- [Try it Yourself »](#)

Orientation: Portrait / Landscape

- Media queries can also be used to change layout of a page depending on the orientation of the browser.
- You can have a set of CSS properties that will only apply when the browser window is wider than its height, a so called "Landscape" orientation

```
@media only screen and (orientation: landscape) {  
  body {  
    background-color: lightblue;  
  }  
}
```

- [Try it Yourself »](#)

Min Width to Max Width

- You can also use the **(max-width: ..)** and **(min-width: ..)** values to set a minimum width and a maximum width.
- For example, when the browser's width is between 600 and 900px, change the appearance of a <div> element

- [Try it Yourself »](#)

```
@media screen and (max-width: 900px) and (min-width: 600px) {  
  div.example {  
    font-size: 50px;  
    padding: 50px;  
    border: 8px solid black;  
    background: yellow;  
  }  
}
```

- **Using an additional value:** In the example below, we add an additional media query to our already existing one using a comma (this will behave like an OR operator)

```
/* When the width is between 600px and 900px OR above 1100px - change the appearance of <div> */
@media screen and (max-width: 900px) and (min-width: 600px), (min-width: 1100px) {
  div.example {
    font-size: 50px;
    padding: 50px;
    border: 8px solid black;
    background: yellow;
  }
}
```

- [Try it Yourself »](#)

Exercise 17

- Please create a web page as shown below. The web page will change the layout based on the size of the viewport. Please upload the web page to the webspace and upload the link to the portal.

